

ELBERT MONUMENT, CO, USA

WMO: 722155

Lat: 39.217N

Lon: 104.633W

Elev: 7060

StdP: 11.31

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 03040

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-0.3	3.2	-9.2	4.3	8.8	-6.1	5.1	17.1	28.9	32.8	26.0	36.6	6.6	200	0.657

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.7	86.1	55.5	82.3	54.8	81.1	54.7	61.2	73.3	59.6	73.2	58.3	72.2	10.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.9	90.2	64.8	54.9	83.7	63.1	54.1	81.3	62.3	30.9	73.0	29.5	73.3	28.6	71.4	67.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.5	23.2	19.6	DB	-13.0	90.1	4.5	3.1	-16.2	92.3	-18.9	94.1	-21.4	95.9	-24.6	98.1	(4)
(5)			WB	-12.7	63.2	4.6	1.2	-16.0	64.0	-18.7	64.7	-21.2	65.3	-24.6	66.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.8	28.6	29.2	36.6	42.0	50.4	61.6	66.7	63.8	58.6	45.6	37.7	27.8
	DBStd	16.32	10.56	10.32	9.64	8.86	8.64	6.87	4.86	5.07	7.12	10.04	10.28	11.15
	HDD50	3320	662	584	421	262	103	5	0	2	18	196	380	687
	HDD65	7195	1127	1002	881	691	455	142	40	79	206	600	820	1152
	CDD50	1790	0	1	5	21	114	352	519	431	276	61	10	0
	CDD65	190	0	0	0	0	1	39	94	43	13	0	0	0
	CDH74	3124	0	0	0	2	79	789	1240	663	331	20	0	0
	CDH80	716	0	0	0	0	4	213	341	120	38	0	0	0
Wind	WSAvg	9.4	8.9	9.7	10.2	11.1	9.8	9.7	8.7	8.5	8.9	9.1	9.5	9.2
Precipitation	PrecAvg	20.5	0.7	0.6	1.5	2.6	2.4	2.3	3.1	3.3	1.4	1.2	0.9	0.7
	PrecMax	30.4	1.5	1.1	3.6	8.0	6.5	5.0	5.6	6.1	2.9	3.0	3.2	2.7
	PrecMin	10.9	0.2	0.1	0.2	0.1	0.7	0.6	1.0	1.5	0.2	0.1	0.1	0.2
	PrecStd	4.2	0.3	0.3	1.0	1.8	1.7	1.2	1.4	1.3	0.8	0.9	0.8	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.2	62.7	68.1	72.3	80.7	90.1	90.8	86.4	84.1	76.9	68.1	58.7
		MCWB	39.2	40.8	43.6	45.4	51.1	54.7	56.2	56.0	55.0	49.9	43.5	39.1
	2%	DB	53.6	54.4	63.1	68.1	76.6	85.8	86.3	82.3	81.0	72.3	63.2	53.6
		MCWB	36.9	36.9	41.3	45.3	50.3	54.0	56.3	55.0	52.8	47.8	41.4	36.7
	5%	DB	48.1	50.3	59.1	63.6	72.3	81.7	83.5	81.1	78.5	69.6	59.1	48.3
		MCWB	35.0	35.7	39.7	43.2	48.9	53.1	55.5	55.1	52.2	46.8	40.1	34.9
	10%	DB	44.6	45.2	54.4	59.5	68.4	79.0	81.3	78.5	75.0	63.5	54.1	43.5
		MCWB	32.7	32.8	38.0	42.1	47.9	53.2	56.3	54.9	52.5	44.8	37.9	31.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.7	41.0	44.3	48.6	54.2	61.4	64.9	63.8	60.4	52.5	44.6	39.4
		MCDB	56.2	58.9	63.2	65.1	72.0	74.8	74.3	72.6	73.4	67.2	64.0	55.4
	2%	WB	37.3	38.0	42.3	46.2	52.0	58.8	61.7	60.3	57.4	50.2	42.4	37.2
		MCDB	50.7	52.5	60.4	62.3	69.2	72.8	73.2	72.1	71.5	66.8	61.0	51.2
	5%	WB	35.1	35.6	40.6	44.7	50.4	56.7	60.2	58.8	55.2	48.0	40.7	34.9
		MCDB	47.5	48.9	57.4	61.0	66.5	72.2	73.7	71.5	69.6	65.0	57.7	47.8
	10%	WB	32.9	33.3	38.5	42.7	49.1	55.2	58.9	57.3	53.7	46.0	38.5	32.5
		MCDB	43.4	44.5	53.1	58.4	64.2	72.3	73.3	70.7	68.6	61.5	53.1	43.4
Mean Daily Temperature Range	5% DB	MDBR	22.7	22.3	23.7	23.8	24.6	27.8	25.7	25.1	26.6	24.9	24.3	22.6
		MCDBR	26.2	27.0	29.6	30.9	31.0	31.9	30.5	29.0	30.2	31.9	29.0	26.9
		MCWBR	14.6	14.1	14.1	13.9	12.4	11.4	9.0	9.5	11.1	13.9	13.9	15.0
	5% WB	MCDBR	24.8	25.4	28.5	29.2	27.8	27.6	24.6	24.6	26.9	28.8	27.6	25.8
		MCWBR	14.3	14.1	14.0	13.7	12.0	12.7	10.7	10.8	11.6	13.1	13.7	15.1
Clear-Sky Solar Irradiance	taub	0.224	0.229	0.249	0.272	0.292	0.302	0.326	0.317	0.285	0.253	0.233	0.225	
	taud	2.647	2.657	2.604	2.547	2.551	2.537	2.509	2.522	2.588	2.667	2.644	2.636	
	Ebn at Noon	315	325	325	320	312	308	299	301	307	311	308	307	
	Edh at Noon	20	23	28	31	32	32	33	32	28	23	20	19	
All-Sky Solar Radiation	RadAvg	836	1139	1529	1846	2052	2307	2208	1993	1708	1269	918	740	
	RadStd	38	65	112	122	156	101	93	61	83	74	50	38	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (28 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+119	+38

Nomenclature: See separate page